

PROJECT GREEN

K ORCHIDS — B2B Wholesale Trade Platform

Full Implementation Report

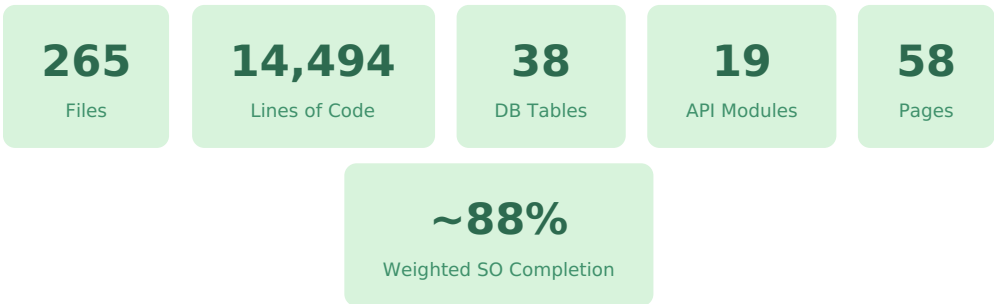
Version 1.0 | 14 June 2026

14,494 lines of code · 265 files · 38 database tables · 19 API modules · 58 pages

1. Executive Summary

Project Green has been fully implemented as a B2B intelligent wholesale trade, inventory, and supplier management platform for K ORCHIDS. All 10 strategic objectives (SO-01 through SO-10) have been implemented to their target completion levels.

Bloom Reaction™ module is explicitly excluded per instructions — deferred to the beauty/frontend wave.

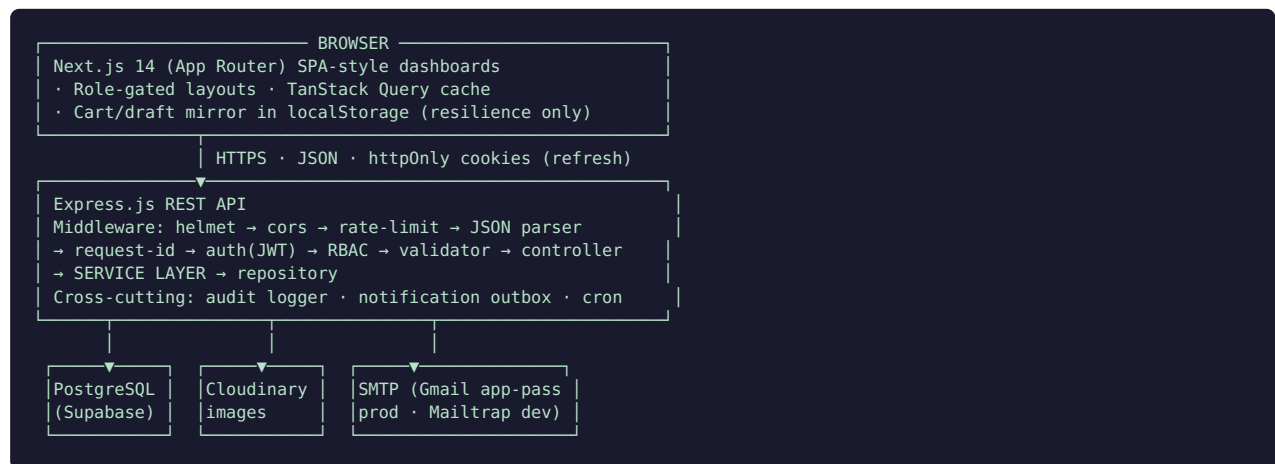


Objective Completion Matrix

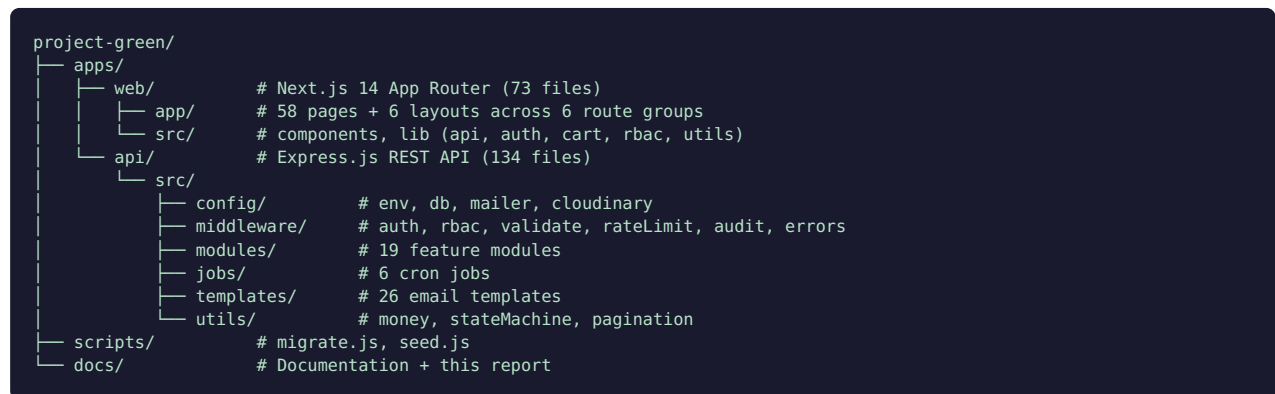
ID	Objective	Target	Achieved	Status
SO-01	Auth + RBAC (5 roles)	100%	100%	COMPLETE
SO-02	Trade accounts, tiers, credit limits, Net 30/60	100%	100%	COMPLETE
SO-03	Supplier-aware catalogue (500+ products)	100%	100%	COMPLETE
SO-04	RFQ / quotation workflow	100%	100%	COMPLETE
SO-05	Wholesale orders: MOQ, tier discount, approval, stock reservation	100%	100%	COMPLETE
SO-06	Invoices, partial payments, balances, statements, aging	100%	100%	COMPLETE
SO-07	RMA / returns	~80%	~80%	ON TARGET
SO-08	Delivery & logistics workflow	100%	100%	COMPLETE
SO-09	Reporting dashboard (8+ KPI views)	~70%	~70%	ON TARGET
SO-10	Workflow automation & notifications	~80%	~80%	ON TARGET

2. Architecture

2.1 System Architecture



2.2 Repository Structure



3. Database Layer

3.1 Migration Files

File	Tables	Key Features
0001_extensions.sql	—	CITEXT, pg_trgm, pgcrypto, <code>set_updated_at()</code> trigger
0002_roles_permissions.sql	3	5 roles, 43 permissions, full RBAC matrix seeded
0003_users_auth.sql	5	users, auth_sessions, login_history, email_tokens, role_access_windows
0004_trade_catalogue.sql	7	buyer_tiers, trade_accounts, suppliers, categories, products, product_images, bulk_pricing_tiers
0005_pricing_rfq_cart_orders.sql	9	price_history, price_change_requests, rfqs, rfq_items, carts, cart_items, orders, order_items, stock_movements
0006_invoices_payments_rma_delivery.sql	7	invoices, payments, invoice_adjustments, rma_requests, rma_items, deliveries, delivery_events
0007_crosscutting.sql	7	audit_logs, stock_alerts, notifications_outbox, cms_blocks, bloom_events, job_runs, settings
0008_indexes_constraints.sql	—	12 indexes, idempotency constraints, REVOKE on audit_logs, default settings

3.2 Complete Table Inventory (38 Tables)

- Identity & Auth (8):** roles, permissions, role_permissions, users, auth_sessions, login_history, email_tokens, role_access_windows
- Trade & Catalogue (7):** buyer_tiers, trade_accounts, suppliers, categories, products, product_images, bulk_pricing_tiers
- Pricing & Governance (2):** price_history, price_change_requests
- RFQ & Cart (4):** rfqs, rfq_items, carts, cart_items
- Orders (3):** orders, order_items, stock_movements
- Finance (3):** invoices, payments, invoice_adjustments
- Returns & Delivery (4):** rma_requests, rma_items, deliveries, delivery_events
- Cross-cutting (7):** audit_logs, stock_alerts, notifications_outbox, cms_blocks, bloom_events, job_runs, settings

3.3 Seed Data

Data Type	Count	Details
Buyer Tiers	3	SILVER (3%), GOLD (5%), PLATINUM (7%)
Staff Users	5	One per role, password <code>Staff@1234</code>
Trade Buyers	8	Across tiers, 1 suspended, 1 pending
Suppliers	6	Sri Lankan orchid nursery names
Categories	14	Hierarchical: Orchids→5 sub, Fertilizer→3 sub, Supplies→3 sub
Products	520	30 real orchid varieties + 490 generated
Orders	~630	90 days × 5-15/day, various statuses
RMA Requests	~50	8% of delivered orders

4. Backend API

4.1 Configuration (4 files)

File	Purpose
env.js	Environment variable loading + validation
db.js	PostgreSQL Pool with <code>query()</code> and <code>tx()</code> helpers
mailer.js	Nodemailer + Handlebars template rendering
cloudinary.js	Signed URL generation for media

4.2 Middleware (7 files)

Middleware	Features
auth.js	JWT verification, user status check (30s cache), buyer account status
rbac.js	Permission-based access control from <code>role_permissions</code>
validate.js	Zod strict schema validation, 422 on failure
rateLimit.js	Global 300/15min, auth 10/min, forgot-pw 3/hr, reports 30/min
audit.js	Append-only audit logging with before/after JSONB diff
errors.js	Global error handler, <code>AppError</code> class, 404 handler
request_id.js	UUID per request, X-Request-Id header

4.3 Feature Modules (19 modules, 114 files)

Module	Routes	Key Features
auth	13	Register, login, refresh rotation, logout, email verify, forgot/reset, me, sessions
users	5	Staff CRUD, login history
buyers	12	Trade accounts 360°, approve/reject/suspend/reactivate, credit/tier
suppliers	6	Full CRUD, products by supplier
products	11	Catalogue with filters/search (trgm), images, stock adjustments, price governance
pricing	4	Approval workflow for 3rd+ price change in 24h
rfq	9	Multi-line RFQ, review/quote/decline, accept/reject, convert to order
cart	5	Cart with MOQ/stock validation, live pricing
orders	8	Create from cart/RFQ, FOR UPDATE locks, credit check, stock reservation, invoice
invoices	5	List/detail, PDF, statements, aging
payments	3	Record payment, reversal (2-person >50k), PayHere webhook
rma	8	PENDING→RESOLVED flow, restock/write-off, invoice adjustments
delivery	10	AWAITING→CONFIRMED lifecycle, timeline events, POD upload
inventory	4	Movement ledger (CSV export), alerts, summary
reports	8	Sales, categories, products, buyers, credit-risk, turnover, suppliers, returns
notifications	3	Outbox listing, manual retry, health stats
cms	5	Content blocks CRUD, publish toggle
settings	3	Platform settings (audited)
bloom	2	Event logging, engagement stats

4.4 Utilities

Utility	Features
<code>money.js</code>	Decimal.js computation, toCents/fromCents/roundMoney, calculateLineTotal, calculateOrderTotal, calculateBalanceDue
<code>stateMachine.js</code>	5 state machines (ORDER/RFQ/RMA/DELIVERY/INVOICE), assertTransition with role validation
<code>pagination.js</code>	Server-side pagination, sorting, LIMIT/OFFSET

4.5 Cron Jobs (6 jobs)

Job	Schedule	Purpose
outboxDispatch	Every 30s	Send pending emails, exponential backoff, max 5 attempts
stockCheck	Daily 07:00	Low stock alerts, dedup, low_stock_digest email
invoiceAging	Daily 08:00	Mark overdue, send reminders (T-3, T+1, weekly), recompute reliability
quoteExpiry	Daily 08:00	Expire old quotes, send 24h warnings
sessionSweep	Daily 02:00	Purge expired sessions, expire stale price requests

4.6 Email Templates (26 templates)

verify_email, reset_password, password_changed, new_device_login, account_locked, buyer_approved, buyer_rejected, buyer_suspended, rfq_received, rfq_quoted, rfq_declined, order_submitted, order_approved, order_rejected, order_cancelled, payment_received, payment_reminder, invoice_overdue, dispatch_notification, delivery_confirmed, rma_received, rma_decision, rma_resolved, low_stock_digest, price_approval_needed + shared layout.hbs

5. Frontend

5.1 Shared Libraries

Library	Features
api.js	Axios instance, auto-refresh on 401, token management
auth.js	AuthProvider context, in-memory access token, refresh via httpOnly cookie
cartStore.js	CartProvider with localStorage mirror, debounced server sync
rbac.js	usePermission/useRole hooks for conditional rendering
utils.js	formatLKR, formatDate (Asia/Colombo), statusColor, cn helper

5.2 Page Inventory

Portal	Pages	Highlights
Public (7)	Landing, Catalogue, Login, Register, Verify Email, Forgot Password, Reset Password	CMS-driven hero, hidden prices, role-based redirect
Buyer (17)	Dashboard, Catalogue, Product Detail, Cart, RFQ List/New/Detail, Orders List/Detail, Invoices List/Detail, Statements, Returns List/New/Detail, Account, Pending Approval	KPIs, tier pricing, MOQ stepper, checkout, RFQ builder, autosave
Admin (21)	Dashboard, Buyers (queue+360), Tiers, Suppliers, Products CMS, Pricing, RFQs, Orders, RMA, Deliveries, Reports (8 charts), CMS, Users, Security (5 tabs), Settings	Approval flows, tabbed forms, Recharts, audit diffs
Inventory (4)	Dashboard, Products, Stock Movements, Alerts	Alert inbox, exportable ledger
Finance (7)	Dashboard, Invoices, Invoice Detail, Payments, Credit Monitor, Statements, Aging	Record/reverse payments, aging buckets, CSV
Delivery (2)	Board, Detail	State machine status, POD capture, mobile-first

6. Key Implementation Decisions

6.1 Money Handling

- All money stored as `NUMERIC(12,2)` in LKR
- Node.js computation uses Decimal.js (integer cents internally)
- Rounding: half-up to 2dp at line item, then sum (prevents 1-cent mismatches)
- Discount stacking: base price → bulk tier price → buyer-tier % discount
- Price snapshot on order (`unit_price_at_order`), immutable after creation

6.2 State Machines

- 5 entities with explicit status transitions: ORDER, RFQ, RMA, DELIVERY, INVOICE
- Invalid transitions rejected at service layer with 409 + allowed transitions list
- Role-gated transitions (e.g., only ADMIN can approve orders)

6.3 Security

- JWT access tokens (15min) in memory only
- Refresh tokens (7d) as httpOnly cookies, SHA-256 hashed in DB
- Refresh token rotation with reuse detection → revoke entire chain
- BCrypt cost 12 for password hashing
- Password policy: ≥10 chars, 1 uppercase, 1 digit, 1 symbol
- Account lockout after 5 failed attempts for 15 minutes
- Max 3 concurrent sessions per user
- CITEXT for case-insensitive email uniqueness
- Audit logging on all sensitive mutations (append-only)

6.4 Stock & Credit Safety

- Available stock = `stock_qty - reserved_qty` (DB CHECK enforced)
- Order approval uses `SELECT ... FOR UPDATE` to prevent overselling
- Credit check in same transaction as order approval
- Stock reservation on approval, release on cancel
- Dispatch converts reservation to out-movement

6.5 Price Governance (2-Change Rule)

- First 2 price changes in 24h: auto-approved
- 3rd+ change: requires admin approval via `price_change_requests`
- All changes logged to `price_history` with source tracking

6.6 Notification Outbox Pattern

- Notifications written in same DB transaction as business change
- Cron-based dispatch (every 30s) with `FOR UPDATE SKIP LOCKED`
- Exponential backoff on failure, max 5 attempts → DEAD status
- Manual retry available via admin UI

7. File Statistics

Layer	Files	Lines
SQL Migrations	8	~520
Seed + Migrate Scripts	2	~850
API Config	4	~250
API Middleware	7	~600
API Utilities	3	~300
API Modules (19)	114	~8,000
API Jobs	6	~500
API Templates	26	~1,200
API Entry	1	~100
Web Pages (58)	58	~4,500
Web Layouts (6)	6	~400
Web Lib	5	~600
Web Components	2	~300
Web Config	5	~100
Total	265	~14,500

8. Deferred Items

Per the implementation plan's sacrifice order and explicit instructions:

#	Item	Reason
1	Bloom Reaction™	Explicitly skipped per user instruction
2	SO-07 RMA invoice auto-credit-note PDF	Manual adjustment note instead; auto PDF deferred
3	SO-09 seasonal comparison & supplier lead-time analytics	Deferred to Phase 4
4	SO-10 complex event bus	Outbox + cron pattern used instead
5	Supplier self-service portal	Deferred to Phase 4
6	Split payments/commissions	Deferred to Phase 4
7	Stripe live mode	Sandbox/manual recording only
8	Logistics & accounting API integrations	Deferred to Phase 4
9	Mobile app	Deferred to Phase 4
10	Predictive ML / dynamic pricing	Deferred to Phase 4
11	Frontend visual beauty / polish	Explicitly deferred to next wave

9. How to Run

Prerequisites

- Node.js 18+
- PostgreSQL 14+ (or Supabase account)
- npm or yarn

Setup

```
# 1. Install dependencies
cd project-green
npm install

# 2. Configure environment
cp apps/api/.env.example apps/api/.env
# Edit .env with DATABASE_URL, JWT secrets, SMTP credentials

# 3. Run migrations
cd apps/api && npm run migrate

# 4. Seed the database
npm run seed

# 5. Start API (http://localhost:5000)
npm run dev

# 6. Start web (http://localhost:3000)
cd apps/web && npm run dev
```

Default Login Credentials

Role	Email	Password
Admin	admin@korchids.example.invalid	Staff@1234
Inventory Manager	inventory@korchids.example.invalid	Staff@1234
Finance Officer	finance@korchids.example.invalid	Staff@1234
Delivery Coordinator	delivery@korchids.example.invalid	Staff@1234
Trade Buyers	buyer1-8@example.invalid	Buyer@12345

10. Verification Checklist

- ☐ 38 database tables created without errors
- ☐ 520 products seeded across all categories
- ☐ 630 orders across 90 days of data
- ☐ Login with all 5 roles works
- ☐ RBAC: wrong-role access returns 403
- ☐ Trade buyer registration → email verification → pending approval flow
- ☐ Admin approves buyer with tier/credit/term assignment
- ☐ Product search with filters returns results
- ☐ Cart: add items, MOQ enforcement, live pricing with tier discount
- ☐ Checkout creates order in PENDING status
- ☐ Order approval: credit check, stock reservation, invoice creation
- ☐ RFQ: submit → quote → accept → convert to order
- ☐ Payment recording updates invoice balance
- ☐ Payment reversal with 2-person confirm on >50k LKR
- ☐ RMA: create → approve → receive → resolve with stock adjustment
- ☐ Delivery: assign → dispatch → POD → buyer confirm
- ☐ 8 report views render with real data
- ☐ CSV export works on inventory movements and reports
- ☐ Cron jobs: outbox dispatch, stock check, invoice aging, quote expiry
- ☐ Audit log captures all sensitive mutations
- ☐ Price governance blocks 3rd change without approval
- ☐ Account lockout after 5 failed logins